



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

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Subject: Programming with Java

Sem: III AY: 2026-27

ASSIGNMENT No: 5

TITLE: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement :

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives:

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory :

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Exercises:

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

Source Code:

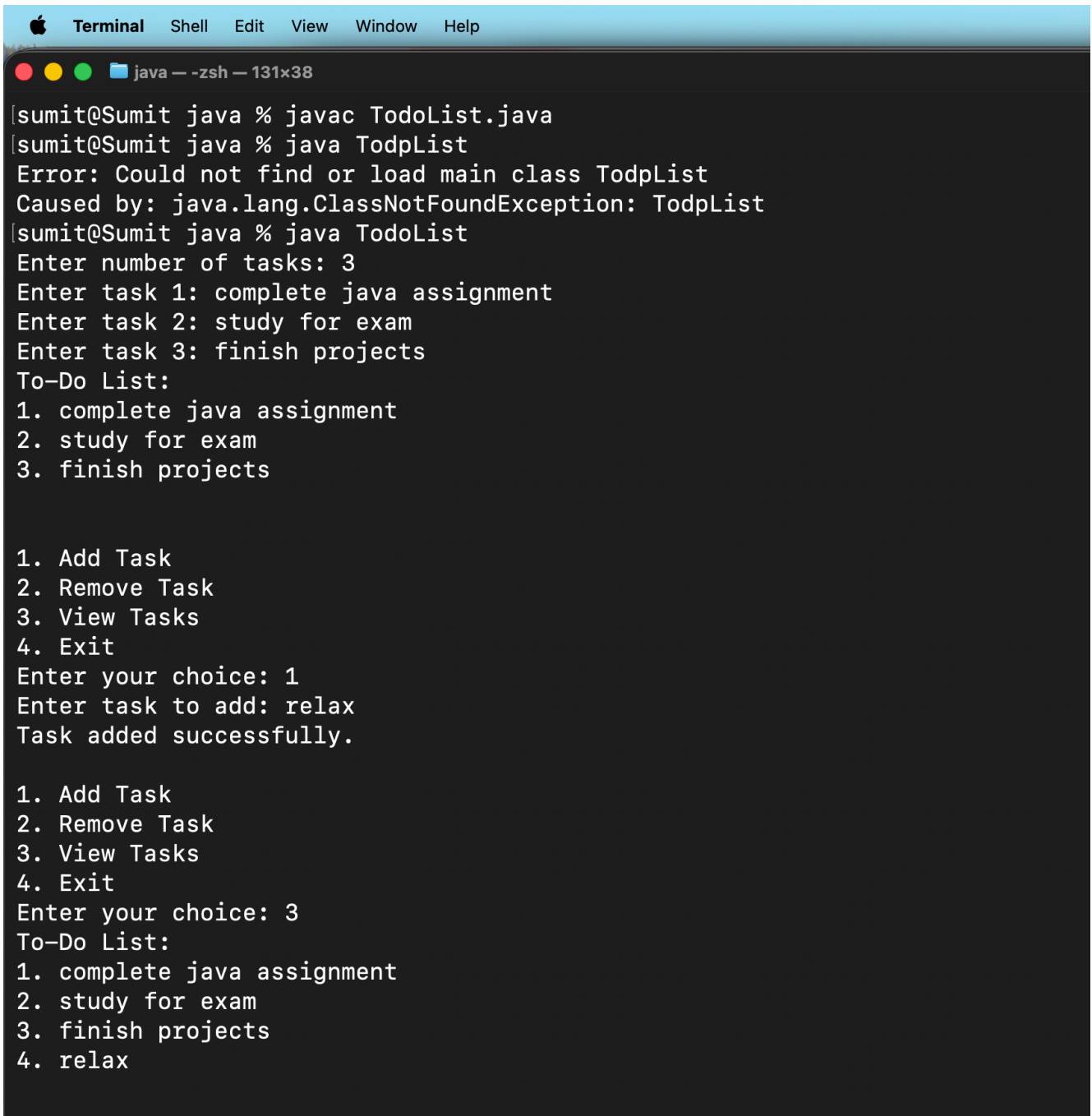
```
1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  public class TodoList
5  {
6      public static void main(String[] args)
7      {
8          ArrayList<String> tasks = new ArrayList<>();
9          Scanner sc = new Scanner(System.in);
10
11          System.out.print("Enter number of tasks: ");
12          int n = sc.nextInt();
13          sc.nextLine(); // Consume newline
14
15          // Store tasks using ArrayList
16          for(int i = 0; i < n; i++)
17          {
18              System.out.print("Enter task " + (i + 1) + ": ");
19              tasks.add(sc.nextLine());
20          }
21
22          // Display tasks using StringBuffer
23          StringBuffer sb = new StringBuffer();
24          sb.append("To-Do List:\n");
25
26          for(int i = 0; i < tasks.size(); i++)
27          {
28              sb.append((i + 1) + ". " + tasks.get(i) + "\n");
29          }
30
31          System.out.println(sb);
32
33          int choice;
34
```

```

35     do
36     {
37         System.out.println("\n1. Add Task");
38         System.out.println("2. Remove Task");
39         System.out.println("3. View Tasks");
40         System.out.println("4. Exit");
41         System.out.print("Enter your choice: ");
42
43         choice = sc.nextInt();
44         sc.nextLine(); // Consume newline
45
46         switch(choice)
47         {
48             case 1:
49                 System.out.print("Enter task to add: ");
50                 String task = sc.nextLine();
51                 tasks.add(task);
52                 System.out.println("Task added successfully.");
53                 break;
54
55             case 2:
56                 System.out.print("Enter task to remove: ");
57                 String remove = sc.nextLine();
58
59                 if(tasks.remove(remove))
60                 {
61                     System.out.println("Task removed successfully.");
62                 }
63                 else
64                 {
65                     System.out.println("Task not found.");
66                 }
67                 break;
68
69             case 3:
70                 StringBuffer sb1 = new StringBuffer();
71                 sb1.append("To-Do List:\n");
72
73                 for(int i = 0; i < tasks.size(); i++)
74                 {
75                     sb1.append((i + 1) + ". " + tasks.get(i) + "\n");
76                 }
77
78                 System.out.println(sb1);
79                 break;
80
81             case 4:
82                 System.out.println("Exiting...");
83                 break;
84
85             default:
86                 System.out.println("Invalid choice. Please try again.");
87         }
88
89     } while(choice != 4);
90
91     sc.close();
92 }
93 }

```

OUTPUT:



```
Terminal  Shell  Edit  View  Window  Help
java — zsh — 131x38

sumit@Sumit java % javac TodoList.java
sumit@Sumit java % java TodpList
Error: Could not find or load main class TodpList
Caused by: java.lang.ClassNotFoundException: TodpList
sumit@Sumit java % java TodoList
Enter number of tasks: 3
Enter task 1: complete java assignment
Enter task 2: study for exam
Enter task 3: finish projects
To-Do List:
1. complete java assignment
2. study for exam
3. finish projects

1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 1
Enter task to add: relax
Task added successfully.

1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 3
To-Do List:
1. complete java assignment
2. study for exam
3. finish projects
4. relax
```

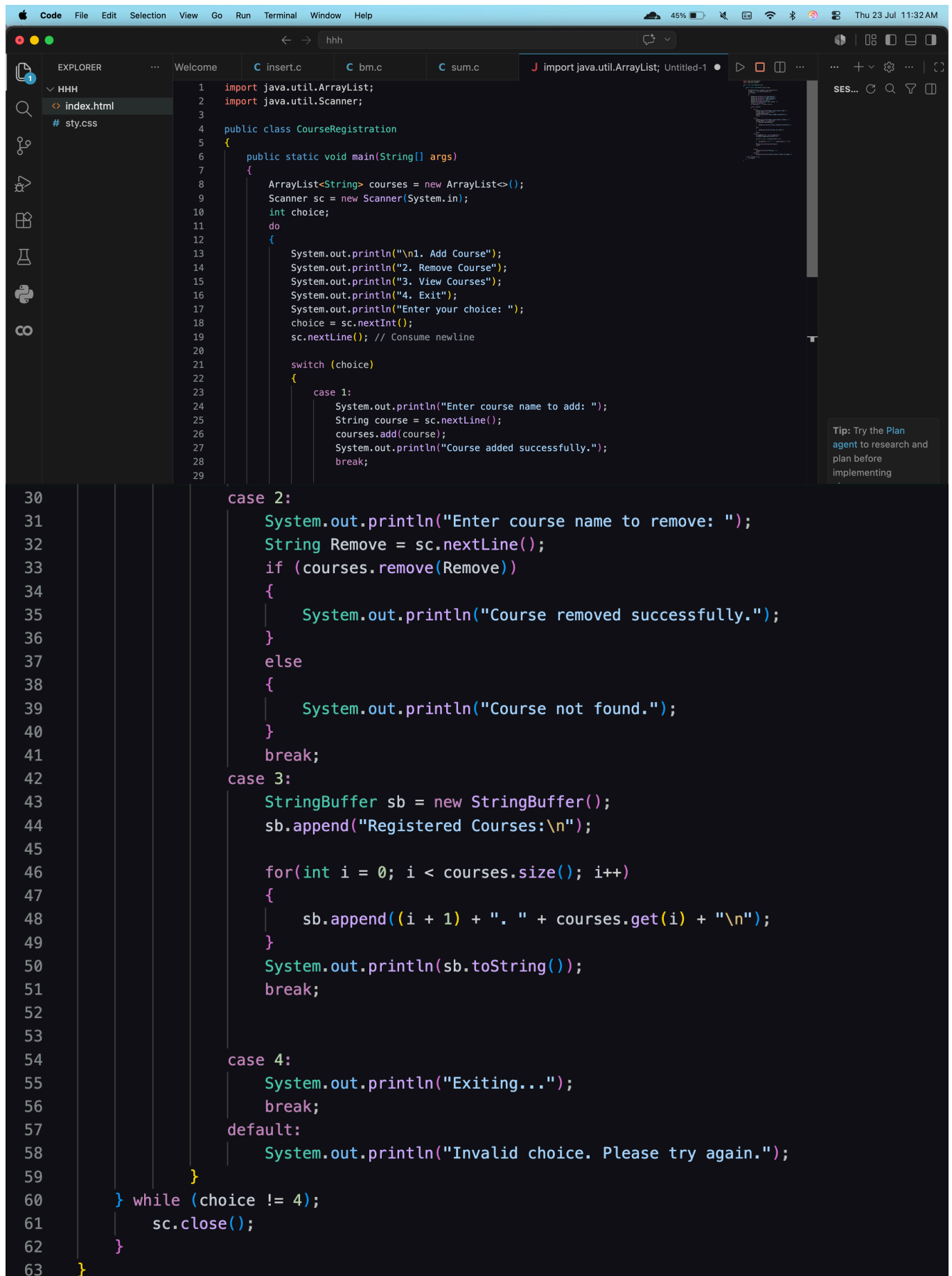
```
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 2
Enter task to remove: relax
Task removed successfully.
```

```
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 3
To-Do List:
1. complete java assignment
2. study for exam
3. finish projects
```

```
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 4
Exiting...
sumit@Sumit java %
```

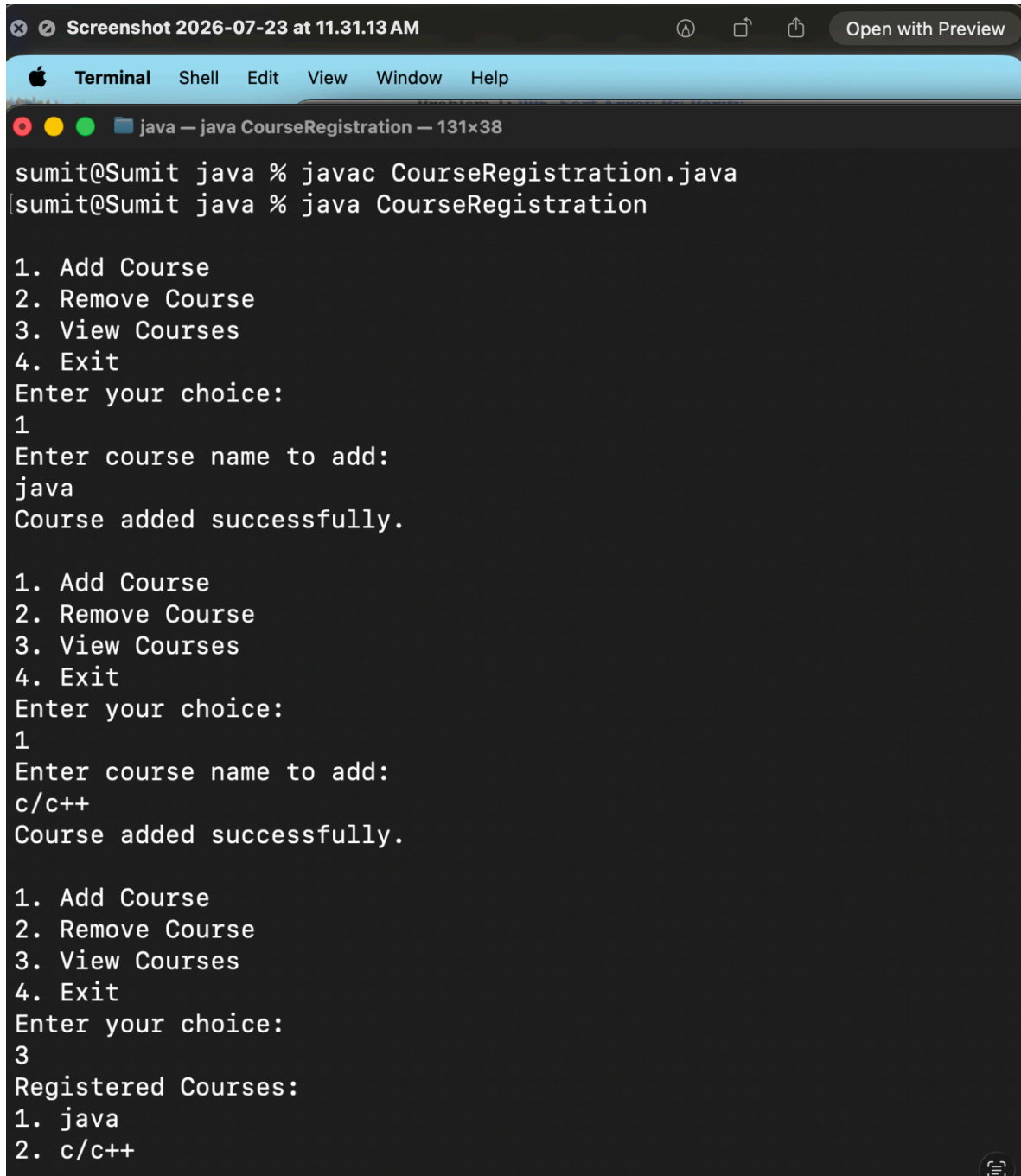
2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

Source Code:



```
1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  public class CourseRegistration
5  {
6      public static void main(String[] args)
7      {
8          ArrayList<String> courses = new ArrayList<>();
9          Scanner sc = new Scanner(System.in);
10         int choice;
11         do
12         {
13             System.out.println("\n1. Add Course");
14             System.out.println("2. Remove Course");
15             System.out.println("3. View Courses");
16             System.out.println("4. Exit");
17             System.out.println("Enter your choice: ");
18             choice = sc.nextInt();
19             sc.nextLine(); // Consume newline
20
21             switch (choice)
22             {
23                 case 1:
24                     System.out.println("Enter course name to add: ");
25                     String course = sc.nextLine();
26                     courses.add(course);
27                     System.out.println("Course added successfully.");
28                     break;
29
30                 case 2:
31                     System.out.println("Enter course name to remove: ");
32                     String Remove = sc.nextLine();
33                     if (courses.remove(Remove))
34                     {
35                         System.out.println("Course removed successfully.");
36                     }
37                     else
38                     {
39                         System.out.println("Course not found.");
40                     }
41                     break;
42                 case 3:
43                     StringBuffer sb = new StringBuffer();
44                     sb.append("Registered Courses:\n");
45
46                     for(int i = 0; i < courses.size(); i++)
47                     {
48                         sb.append((i + 1) + ". " + courses.get(i) + "\n");
49                     }
50                     System.out.println(sb.toString());
51                     break;
52
53                 case 4:
54                     System.out.println("Exiting...");
55                     break;
56                 default:
57                     System.out.println("Invalid choice. Please try again.");
58             }
59         } while (choice != 4);
60         sc.close();
61     }
62 }
63 }
```


Output:



```

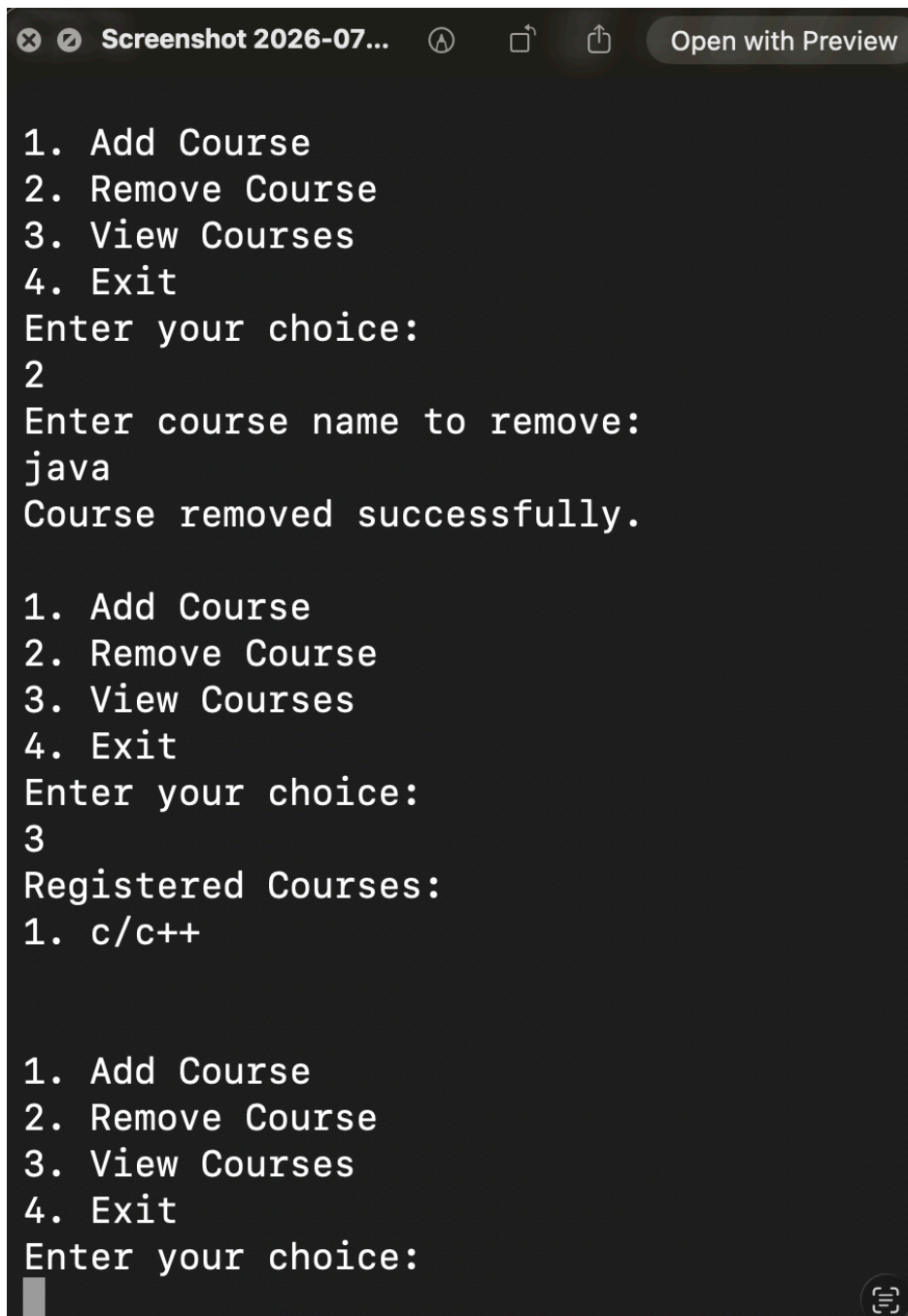
Screenshot 2026-07-23 at 11.31.13 AM
Terminal Shell Edit View Window Help
java — java CourseRegistration — 131x38
sumit@Sumit java % javac CourseRegistration.java
sumit@Sumit java % java CourseRegistration

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter your choice:
1
Enter course name to add:
java
Course added successfully.

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter your choice:
1
Enter course name to add:
c/c++
Course added successfully.

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter your choice:
3
Registered Courses:
1. java
2. c/c++

```



```
Screenshot 2026-07... Open with Preview

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter your choice:
2
Enter course name to remove:
java
Course removed successfully.

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter your choice:
3
Registered Courses:
1. c/c++

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter your choice:
```

Conclusion:

This experiment demonstrated the practical implementation of ArrayList, Vector, and StringBuffer in Java. ArrayList and Vector were used for dynamic data storage and management, while StringBuffer was used for efficient string manipulation. The developed applications successfully performed add, remove, and display operations, improving the understanding of Java Collections Framework and mutable string handling in real-world programming.